

Task 5: Button Interrupt

5_Button_Interrupt.ino

```
1  #define BTNPIN 23
2  #define LEDPIN 22
3  volatile bool buttonPressed = false; // flag set by ISR
4
5  // Interrupt Service Routine (ISR)
6  void IRAM_ATTR handleButtonInterrupt() {
7      buttonPressed = true; // set flag
8  }
9
10 void setup() {
11     Serial.begin(115200);
12
13     pinMode(BTNPIN, INPUT_PULLUP); // button between GND and GPIO23
14     pinMode(LEDPIN, OUTPUT);
15     // attach interrupt to BTNPIN, trigger on falling edge (button press)
16     attachInterrupt(digitalPinToInterrupt(BTNPIN), handleButtonInterrupt, FALLING);
17 }
18
19 void loop() {
20     if (buttonPressed) {
21         buttonPressed = false; // clear flag
22
23         // do your action here
24         Serial.println("Button Pressed!");
25         digitalWrite(LEDPIN, HIGH);
26     }
27 }
28 }
29
```